PLATFORM REPORT · 2026

NexaAssist

Centralized AI-powered business operations

NexaAssist is a **multi-tenant business operations platform**. Each organization (tenant) runs in its own secure workspace with role-based access control. Teams manage **scheduling, inventory, AI-assisted workflows, marketing campaigns, SEO audits, and WhatsApp messaging** from a single dashboard—all modules share the same data model, permissions, and audit trails.

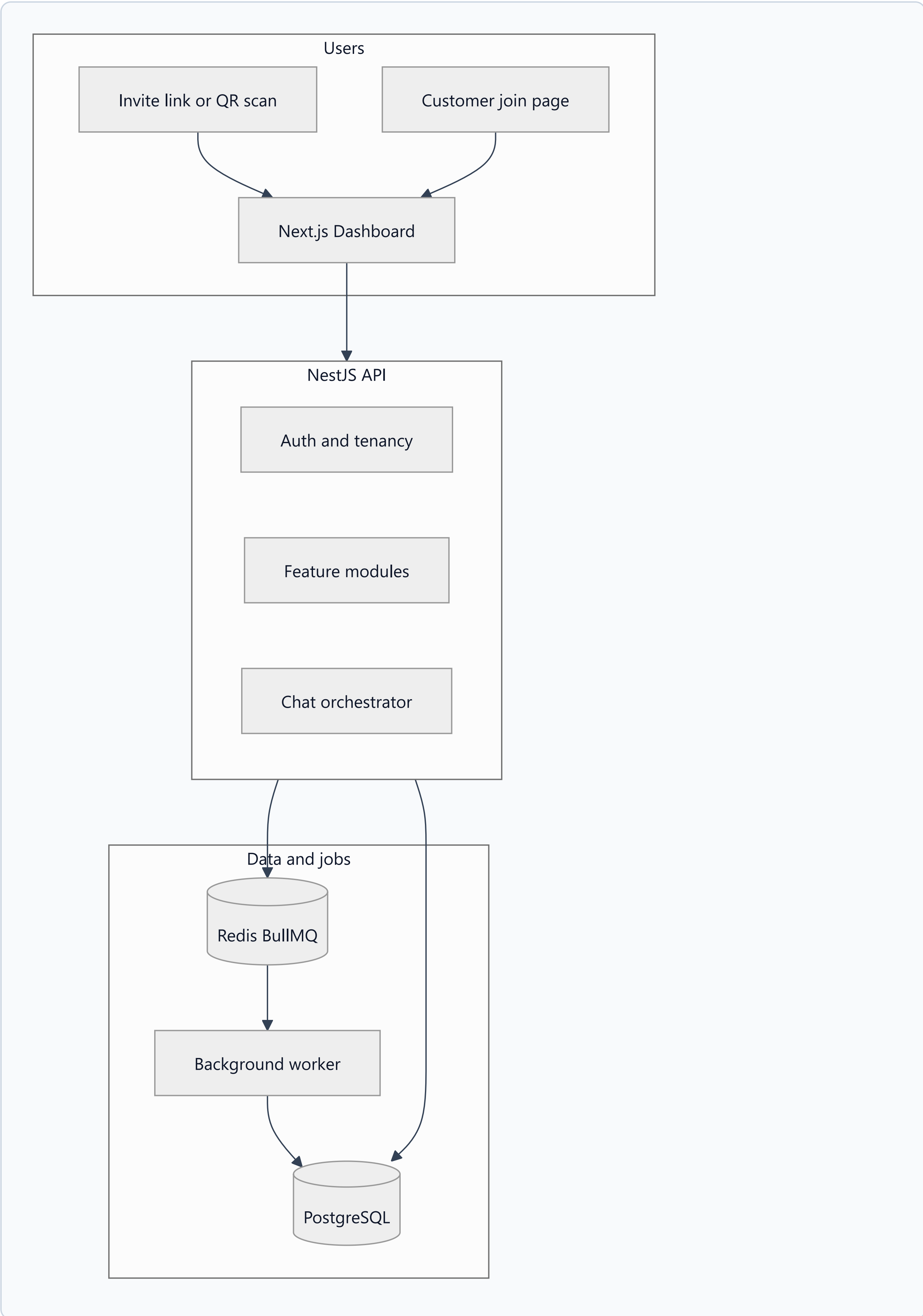
Appointments Scheduling & calendar	**Inventory** Stock & restock	**AI Assistant** 13 operational tools
Campaigns AI copy & posters	**SEO Audit** Crawl & recommendations	**WhatsApp** Bulk messaging

CHAPTER 01

Platform at a glance

> Every feature follows one architecture: the web dashboard calls a secure API, which persists data in PostgreSQL and delegates long-running work to background processors.

The **Next.js dashboard** is what staff and customers use day to day. It communicates with a **NestJS API** (all routes under `/api`), which reads and writes **PostgreSQL**. Work that takes time—appointment reminder emails, SEO site scans, WhatsApp batch sends, campaign execution, and analytics rollups—is queued in **Redis** (BullMQ) and processed by a dedicated **background worker** so the API stays fast and responsive.


Users
Invite link or QR scan
Customer join page
Next.js Dashboard
NestJS API
Auth and tenancy
Feature modules
Chat orchestrator
Data and jobs
Redis BullMQ
Background worker
PostgreSQL

Foundation: organizations, users, and access

Organizations (tenants)

Each business is a **tenant** with its own name, URL slug, and configuration. All appointments, inventory records, campaigns, SEO projects, chat sessions, and analytics are **scoped to that tenant**. Users belonging to one organization cannot view or modify another organization's data.

Users and roles

Access is controlled through **roles**. Each role receives a defined set of **permission codes** (for example `appointments:read`, `inventory:write`, `chat:use`). The dashboard only displays navigation items and action buttons that the signed-in user is permitted to use.

ROLE	TYPICAL RESPONSIBILITY
SUPER_ADMIN	Platform-wide administration across all tenants
TENANT_ADMIN	Organization owner; full tenant setup, users, invites, and modules
DOCTOR	Clinical staff; appointments, availability, inventory consumption
RECEPTIONIST	Front desk; booking, calendar, customer-facing scheduling
INVENTORY_MANAGER	Stock control, alerts, restock approvals
STAFF	General operational access
CUSTOMER	End client; books appointments after joining an organization

User codes

Every account receives a human-readable **user code** for quick identification in the UI and in assistant conversations—for example `TA-acme-clinic-001` for a tenant administrator, or `CU-INV-001` for a customer not yet linked to an organization.

Sessions and security

Users authenticate with email and password. The API issues a short-lived **JWT access token** and a longer-lived **refresh token**. The dashboard stores these securely and refreshes access automatically. When a session expires, the user must sign in again. Every API request is validated against the user's tenant and permission list.

CHAPTER 03

Getting people into the workspace

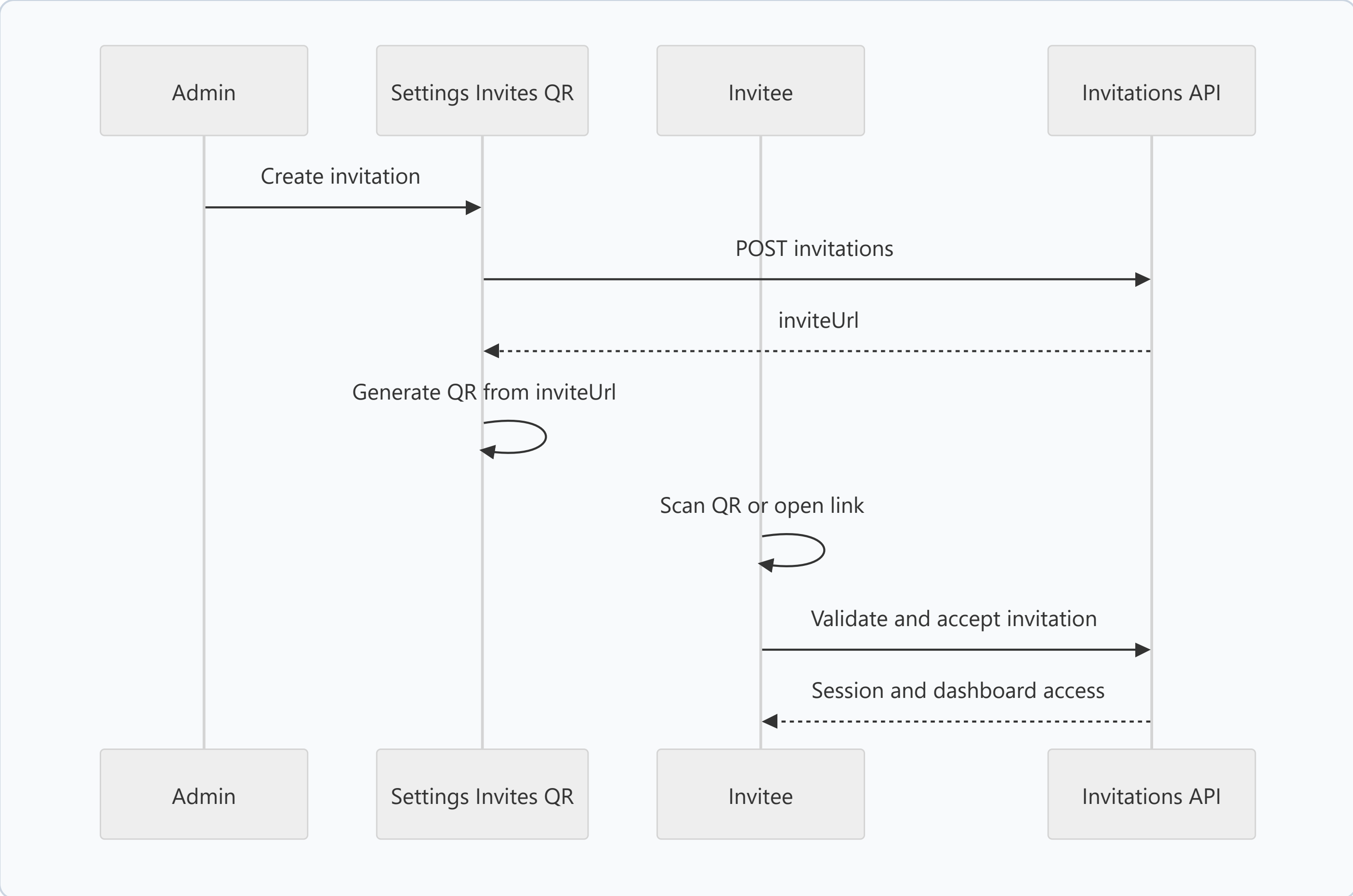
> NexaAssist supports multiple onboarding paths so business owners, staff, and customers each enter the platform through the correct door—with optional QR codes for frictionless staff invites.

Onboarding paths

PATH	WHO USES IT	WHAT HAPPENS
Register organization	New business administrator	Creates a new tenant and a **TENANT_ADMIN** account. The admin lands on the dashboard to configure service types, availability, and staff invites.
Register customer	End client	Creates a **CUSTOMER** account **without** a tenant. The customer must request access to an organization before booking or using most modules.
Join organization	Customer	Submits a **join request** to a chosen organization. A tenant admin **approves** or rejects on the Join Requests screen. Once approved, the customer is linked and can book appointments.
Invite staff (link + QR)	Tenant admin	Admin creates an invitation under **Settings → Invites**. The system returns a unique **invite URL** and the dashboard renders a **QR code** in the browser. The invitee scans or opens the link, accepts, and is signed in.
Standalone join page	Customer	Public `/join` page combines registration and organization access request in a single flow.

Invite flow with QR code

When an administrator creates an invitation, the API returns a unique tokenized URL. The settings screen generates a **QR code** from that URL client-side (no image stored on the server). The invitee scans the code or taps the link, lands on `/invite/[token]`, validates the invitation, accepts it, and receives an authenticated session with the assigned role.

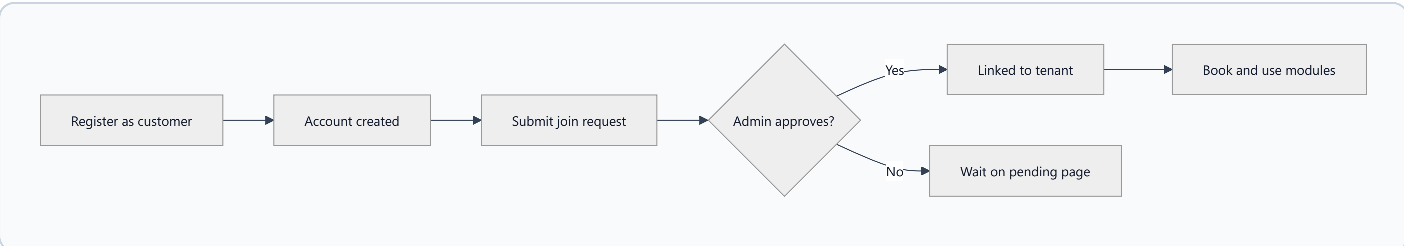


> **QR codes are for staff and customer invites only**—not for appointment check-in at the door. Each appointment has its own reference code (**APT-...**) shown in lists, detail views, and reminder emails.

Customer without an organization

Customers who register without a tenant are directed to **Organization access** (`/dashboard/pending-tenant`) until an administrator approves their join request. Scheduling, inventory, and most other modules remain hidden until the customer is linked.

Customer registration sequence



CHAPTER 04

Appointments & scheduling

What it is

End-to-end **appointment lifecycle management** for organizations with multiple staff members: define services, configure availability, book time slots, confirm or reject requests, reschedule, mark complete, and send **automated email reminders** for confirmed bookings.

Dashboard screens

SCREEN	PURPOSE
Appointments	List and filter all appointments; open detail for actions and history
Book Appointment	Guided booking wizard (service → staff → day → slot)
Calendar	Day, week, and month views with appointments and availability overlays
Availability	Recurring weekly schedules, one-off slots, and leave/blocked periods
Service Types	Services with duration and color; duration drives slot length in booking

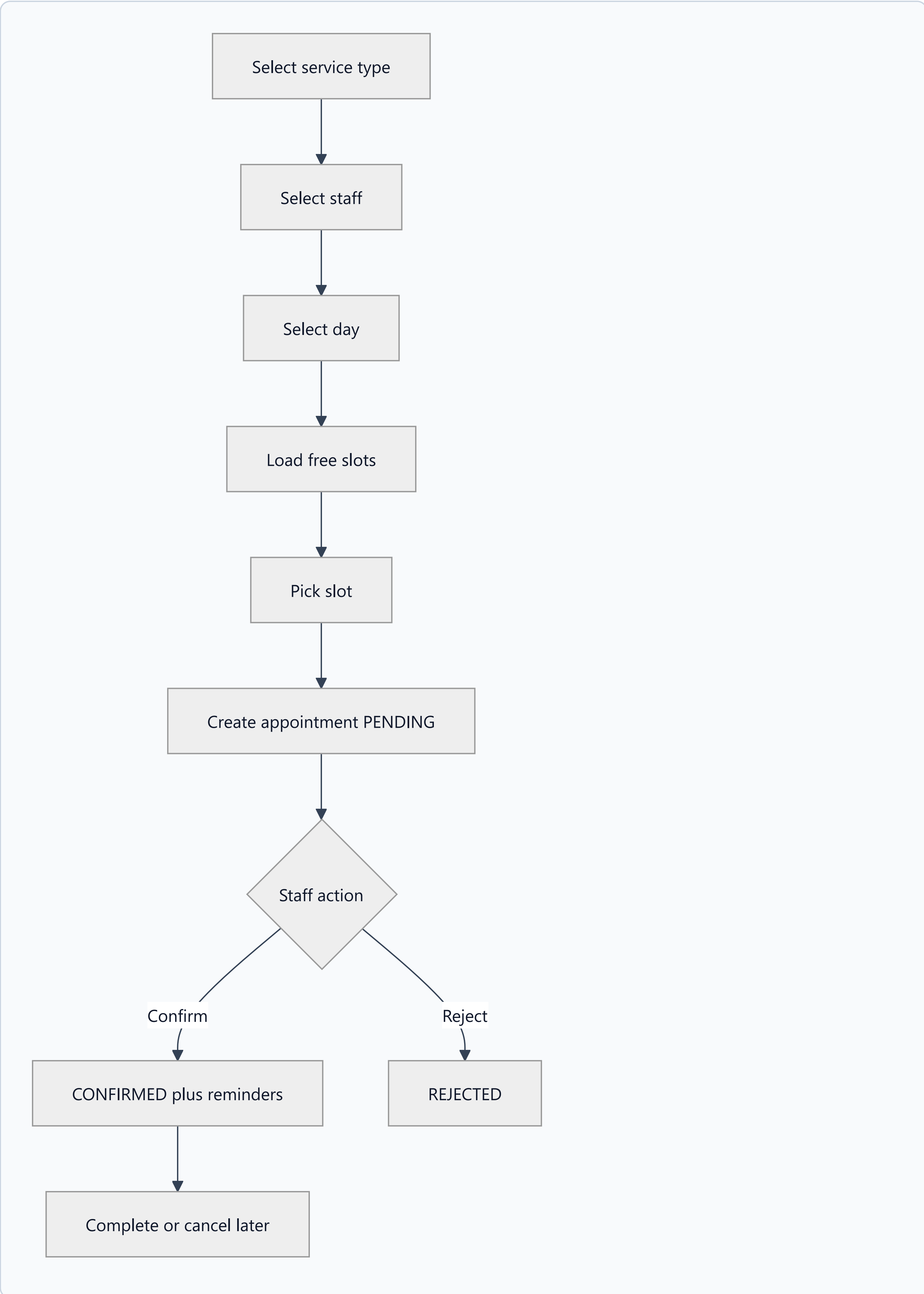
Key capabilities

- **Service types** — name, duration in minutes, display color; used when calculating free slots
- **Staff availability** — recurring weekly rules plus individual availability slots; **blocked** windows for leave or unavailability
- **Free slots** — the system computes bookable time windows by subtracting existing appointments and blocked periods from available hours
- **Appointment codes** — every booking receives a unique `APT-…` reference for staff and customers
- **Status lifecycle** — `PENDING` → `CONFIRMED` or `REJECTED` → `COMPLETED`, `CANCELLED`, `RESCHEDULED`, or `NO_SHOW`
- **Overlap protection** — the same staff member cannot be double-booked for overlapping times
- **History** — full audit trail of status changes and notes
- **Reminders** — after confirmation, background jobs schedule and send reminder emails

How booking works (step by step)

1. User selects a **service type** (determines appointment duration).
2. User selects **bookable staff** for that service.

3. Reception or staff optionally picks a **customer**; customers book for themselves.
4. User picks a **day**; the UI loads **free slots**, existing appointments, and blocked time for that day.
5. User selects a slot and submits → appointment is created as **PENDING**.
6. Authorized staff **confirm** or **reject**; they may later cancel, reschedule, or mark complete.


Select service type
Select staff
Select day
Load free slots
Pick slot
Create appointment PENDING
Staff action
Confirm
Reject
CONFIRMED plus reminders
REJECTED
Complete or cancel later

Permissions (scheduling)

PERMISSION	ALLOWS
`appointments:read`	View appointments and calendar
`appointments:create`	Book new appointments
`appointments:update`	Confirm, reschedule
`appointments:cancel`	Cancel appointments
`calendar:read`	Calendar day/week/month views
`availability:read` / `write`	View or edit availability and leave
`service-types:read` / `write`	View or manage service types

CHAPTER 05

Inventory

What it is

Tenant-scoped stock control for operational teams: maintain a catalog of items and categories, record every stock movement, receive automatic threshold alerts, and run a formal **restock request** workflow with manager approval before stock is received.

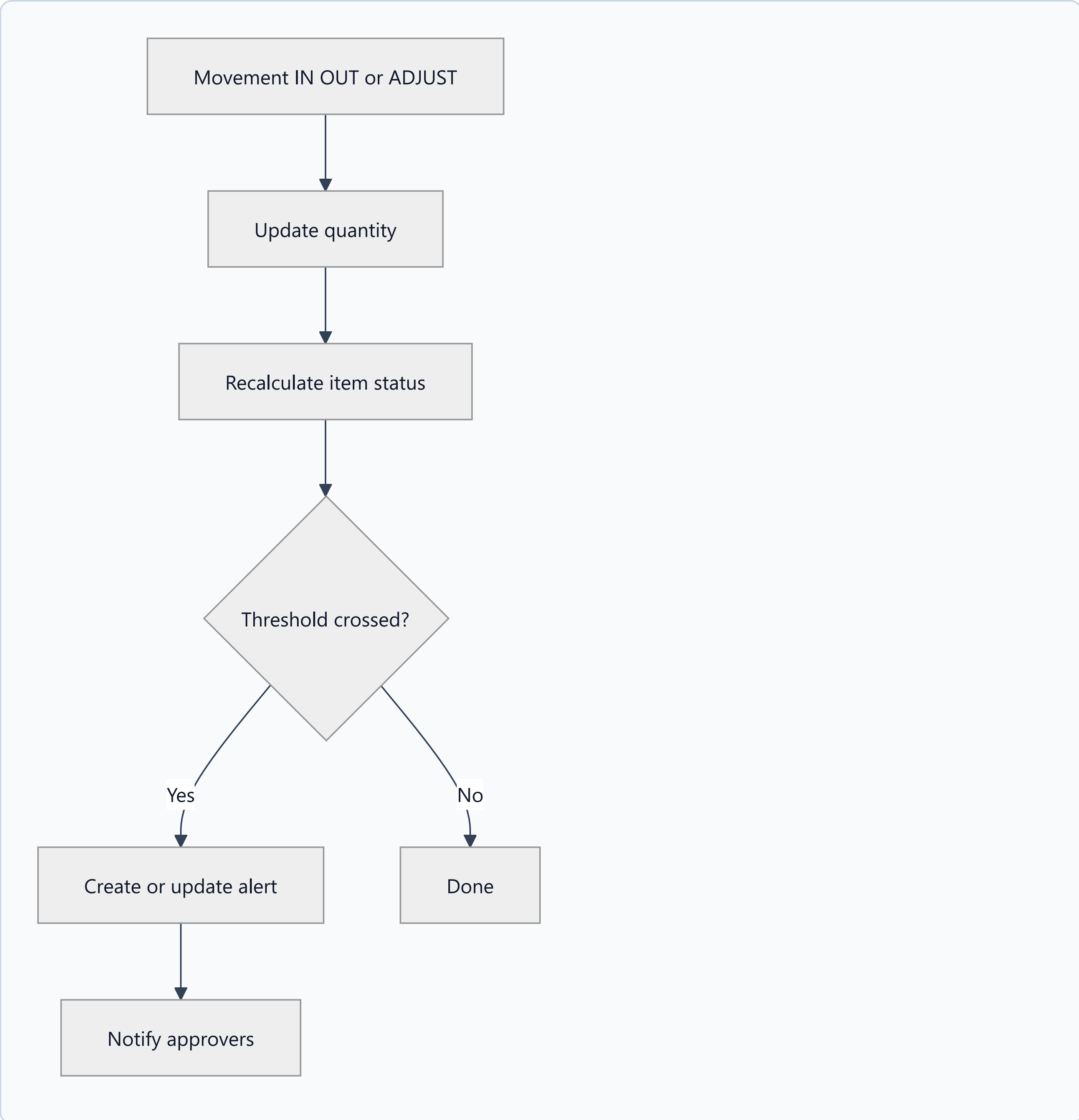
Dashboard screens

SCREEN	PURPOSE
Inventory overview	Metrics, stock health, recent movements, active alerts, pending requests
Items	Search, filter, create items; open detail for stock actions
Categories	Organize the item catalog
Movements	Complete audit log of stock changes
Alerts	Low stock, out of stock, and overstock notifications
Requests	Restock request queue and approval actions

Key capabilities

- **Items** — SKU unique per tenant, on-hand quantity, reserved quantity, available quantity, minimum threshold
- **Status** — automatically derived (active, low stock, out of stock, and related states)
- **Movements** — **IN** (receive), **OUT** (consume), **ADJUSTMENT** (set absolute quantity)
- **Alerts** — generated when thresholds are crossed; staff acknowledge, resolve, or mark as read
- **Restock requests** — submit → manager approves or rejects → fulfill records stock IN and closes the request
- **Dashboard metrics** — total items, low-stock count, critical alerts, movements today, pending approvals

How stock changes propagate



Restock request workflow

1. A user with `inventory:request` creates a **pending** request specifying item and quantity.
2. Users with `inventory:approve` see **all** requests; others see only their own submissions.
3. An approver **approves** or **rejects** with optional notes.
4. On **fulfill**, the system posts an **IN** movement linked to the approved request and marks the request complete.

Permissions (inventory)

PERMISSION	ALLOWS
`inventory:read`	View dashboard, items, movements
`inventory:write`	Categories, items, stock IN
`inventory:consume`	Stock OUT (consumption)
`inventory:adjust`	Quantity adjustments
`inventory:request`	Submit restock requests
`inventory:approve`	Approve, reject, fulfill; view all requests
`inventory:alerts`	Acknowledge and resolve alerts

CHAPTER 06

AI assistant (chatbot)

What it is

The **NexaAssist Assistant** is a tenant-scoped conversational interface that answers operational questions and **executes real actions**—booking appointments, checking stock, creating restock requests, sending invites—using the same business rules, validation, and permissions as the rest of the dashboard. It is not a generic external chatbot; it is wired directly into your organization's data.

Where to use it

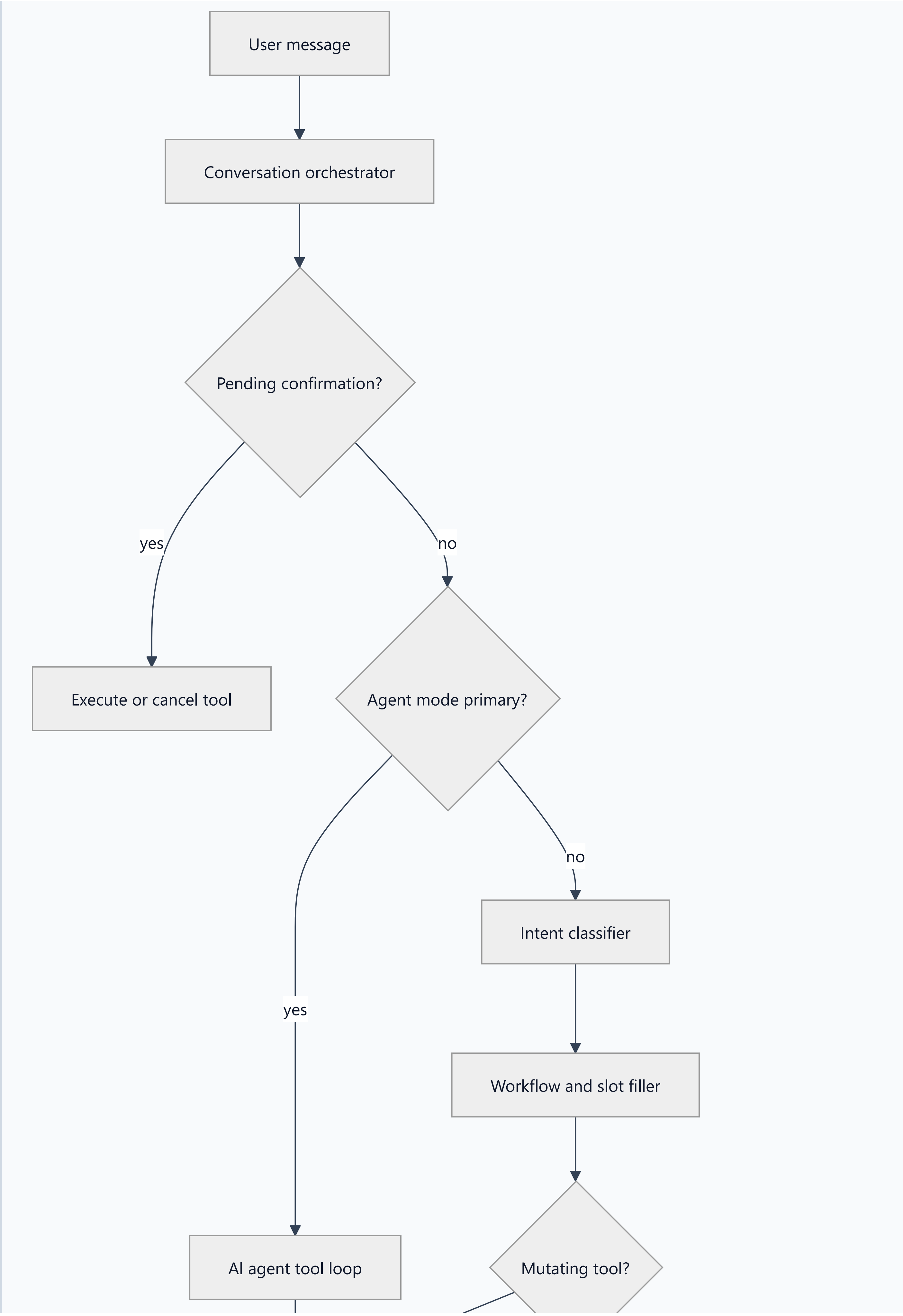
Dashboard → Assistant provides chat sessions, full message history, interactive UI cards (slot pickers, booking summaries, inventory search results, confirmation dialogs), and contextual quick-action suggestions.

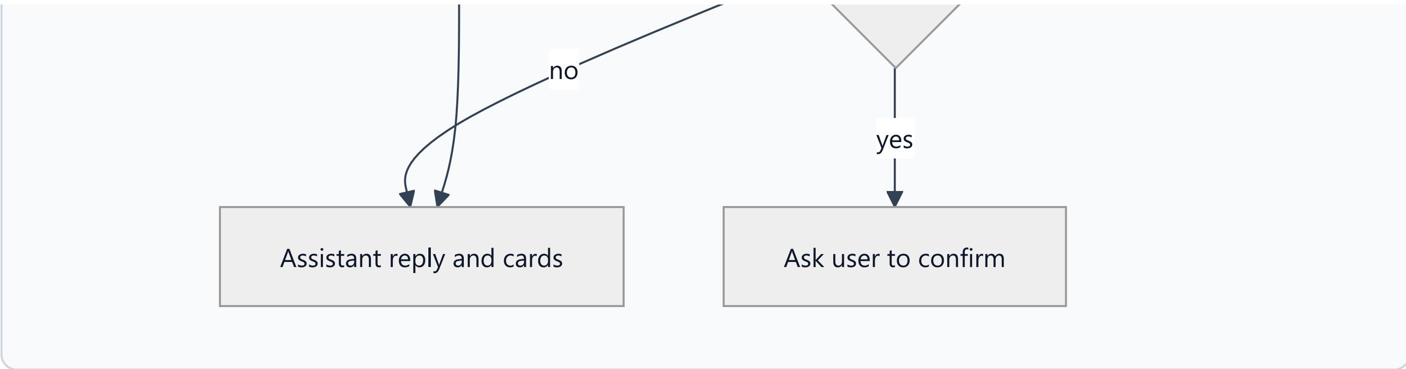
Assistant tools (13)

Each tool is available only when the signed-in user's role includes the required underlying permission:

TOOL	WHAT IT DOES
`listBookableStaff`	List staff available for booking
`getFreeSlots`	Return free slots for a staff member and time range
`countAppointments`	Count appointments on a given date
`listTodayAppointments`	Summarize today's appointments
`createAppointment`	Create a new appointment (requires confirmation)
`confirmAppointment`	Confirm a pending appointment
`cancelAppointment`	Cancel by appointment ID
`rescheduleAppointment`	Move appointment to new start/end times
`searchInventory`	Find items by name or SKU
`createInventoryItem`	Add a new catalog item
`createRestockRequest`	Submit a restock request
`createUser`	Create a tenant user (admin)
`createInvitation`	Create a staff or customer invite

How a message is processed


User message
Conversation orchestrator
Pending confirmation?
yes
no
Execute or cancel tool
Agent mode primary?
no
Intent classifier
yes
Workflow and slot filler
AI agent tool loop
Mutating tool?



1. User sends a message in an active **chat session**.
2. The **orchestrator** persists the message and checks whether a prior step is awaiting yes/no **confirmation**.
3. In **primary agent mode**, an AI agent runs a multi-step **tool loop** to fetch data and propose actions.
4. Otherwise an **intent classifier** routes the message into structured **workflows** (for example “book appointment”) that collect required slots step by step.
5. **Read-only** tools execute immediately and return structured cards in the chat UI.
6. **Mutating** tools require explicit user confirmation before any write occurs.
7. The assistant response is saved with metadata for cards, follow-up suggestions, and audit logging.

Guardrails

- **Role-based access** — tools unavailable to a role are never invoked.
- **Explicit confirmation** — creates, cancels, reschedules, inventory writes, and invites need user approval.
- **Tenant scope** — all scheduling and inventory operations respect the user’s organization.
- **Audit trail** — tool calls and outcomes are logged for accountability.

PERMISSION	ALLOWS
`chat:use`	Open Assistant and send messages

CHAPTER 07

Campaigns

What it is

Marketing campaigns managed inside the tenant: define objectives and audience, set budget, run an approval workflow, generate **AI marketing copy and poster images**, schedule execution, and review per-campaign analytics.

Where to use it

Dashboard → Campaigns — create and edit campaigns, request approval, schedule or execute, and view generated assets and performance metrics.

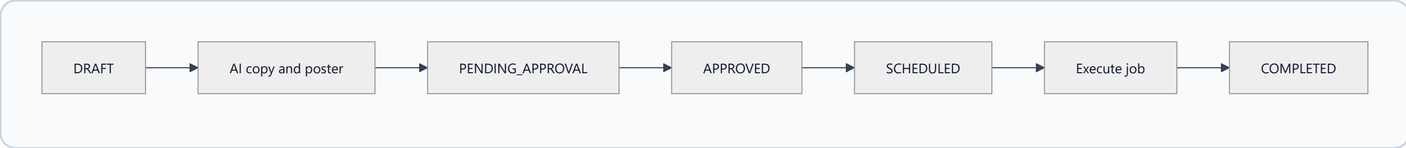
Campaign status lifecycle

STATUS	MEANING
`DRAFT`	Campaign is being edited
`PENDING_APPROVAL`	Submitted; awaiting manager decision
`APPROVED`	Approved and ready to schedule
`SCHEDULED`	Set to run at a future time
`ACTIVE`	Currently running
`COMPLETED`	Finished; analytics recorded
`PAUSED` / `REJECTED` / `ARCHIVED`	Stopped or archived states

How campaigns work

1. **Create** a campaign with objective, audience, and content (starts as draft).
2. Optionally generate **AI copy** and a **poster image** stored as campaign assets.
3. **Request approval** — approver accepts, rejects, or requests revision.
4. **Schedule** or **execute** the campaign.
5. A **background job** runs execution, updates status to completed, and records impressions and analytics.

6 **WhatsApp batches** may reference the same `campaignId` for coordinated outreach.



Permissions (campaigns)

PERMISSION	ALLOWS
`campaigns:read`	View campaigns
`campaigns:write`	Create and edit
`campaigns:approve`	Approve or reject
`campaigns:send`	Schedule and execute

CHAPTER 08

SEO audit

What it is

Website SEO auditing built into the platform: crawl pages, apply rule-based checks, estimate scores, list issues, and enrich findings with **AI-written recommendations** and executive summaries—saved per tenant as projects and scans.

Where to use it

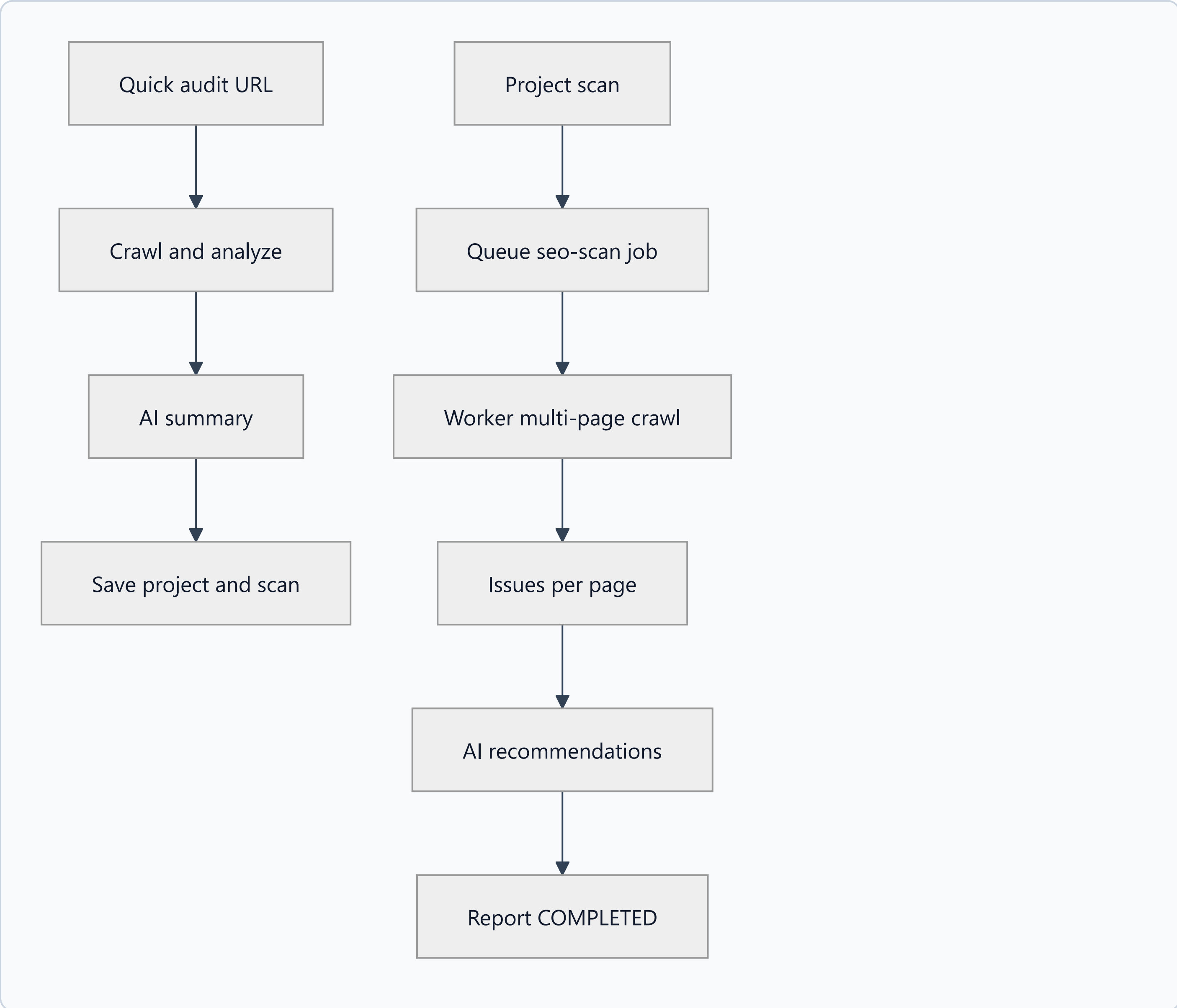
Dashboard → SEO Audit — run an instant single-URL audit, manage **projects** (domains), trigger multi-page **scans**, compare scan results, and download reports.

Quick audit (runs immediately)

1. User enters a **URL**.
2. System **crawls** the page and runs the analyzer (titles, meta tags, headings, links, performance estimates, and more).
3. **AI** produces a narrative summary: strengths, fixes, and grouped categories.
4. Results are **persisted** as a project, scan, page record, and issue list.

Project scan (runs in background)

1. User creates or opens an **SEO project** for a site or domain.
2. User triggers **Scan** — status moves to queued.
3. The **worker** crawls to configured depth, discovers linked pages, and analyzes each.
4. Pages with issues receive **AI recommendations** where applicable.
5. Scan completes; a **report** summarizes scores and issue counts.



Permissions (SEO)

PERMISSION	ALLOWS
`seo:read`	View projects, scans, reports; run quick audit
`seo:write`	Create projects and trigger multi-page scans

CHAPTER 09

Supporting modules

WhatsApp

Dashboard → WhatsApp provides message **templates** with `{{variable}}` placeholders, **batches** of recipients (optionally tied to a campaign), and queued delivery. A worker sends messages in chunks of up to **50 per step** and updates delivery status. When Meta Cloud API credentials are configured, messages go through WhatsApp; otherwise the platform performs a **simulated send** for demonstration. Failed messages can be **retried**; webhooks update delivery state.

Analytics

Dashboard → Analytics shows overview **KPIs** (appointments, users, active campaigns, SEO projects) and dedicated views for appointments, inventory, campaigns, **chatbot usage** (sessions and tool calls), and SEO. An **aggregation worker** rolls up daily metrics and snapshots; administrators can trigger manual aggregation.

Notifications

Dashboard → Notifications and the header bell deliver in-app alerts from **domain events**: appointment changes, inventory alerts, join requests, and system notices.

Operations

Dashboard → Operations exposes **queue health** and **failed job** lists across all BullMQ queues (campaigns, WhatsApp, SEO, analytics, appointments, inventory, and others). Operators can **retry** individual failed jobs.

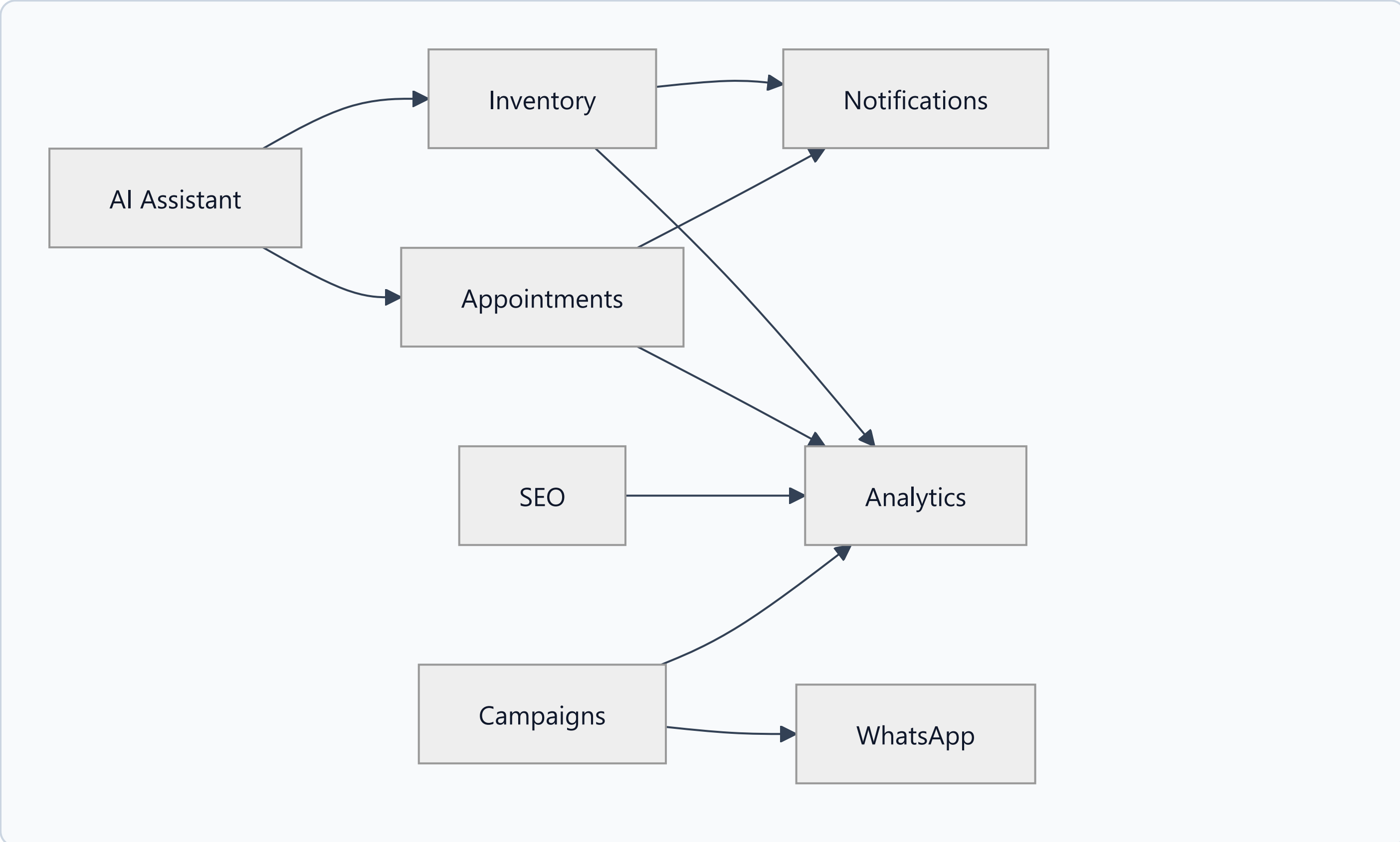
Users, settings, and modules

SCREEN	PURPOSE
Users	List and manage tenant user accounts
Join Requests	Approve or reject customer organization access
Settings	Organization profile, invitations, **QR codes** for invite links
Modules	Tenant module configuration

CHAPTER 10

How modules connect

Modules are logically separate but share data and operational workflows across the platform.



FROM	TO	CONNECTION
Assistant	Appointments, Inventory	Books, confirms, and searches stock via the same APIs as the UI
Campaigns	WhatsApp	Batches can reference a campaign ID
Campaigns	Analytics	Impressions and campaign performance metrics
SEO	Analytics	Project and scan KPIs
Appointments	Notifications, Analytics	Events, reminders, scheduling metrics
Inventory	Notifications, Analytics	Alerts, restock events, stock metrics

Sidebar navigation groups: Scheduling · Operations · Assistant · Marketing & Growth · Admin · System

CHAPTER 11

Dashboard map

Routes appear only when the user holds the required permission and (for customers) is linked to an organization.

Primary routes

GROUP	ROUTE	PERMISSION (TYPICAL)
Home	`/dashboard`	Signed in
Scheduling	`/dashboard/appointments`	`appointments:read`
Scheduling	`/dashboard/calendar`	`calendar:read`
Scheduling	`/dashboard/availability`	`availability:read`
Scheduling	`/dashboard/service-types`	`service-types:read`
Scheduling	`/dashboard/booking`	`appointments:create`
Operations	`/dashboard/inventory`	`inventory:read`
Operations	`/dashboard/operations`	`operations:read`
Assistant	`/dashboard/assistant`	`chat:use`
Marketing	`/dashboard/campaigns`	`campaigns:read`
Marketing	`/dashboard/whatsapp`	`whatsapp:read`
Marketing	`/dashboard/seo`	`seo:read`
Marketing	`/dashboard/analytics`	`analytics:read`
Admin	`/dashboard/users`	`users:read`
Admin	`/dashboard/join-requests`	`join-requests:manage`
Admin	`/dashboard/notifications`	`notifications:read`
System	`/dashboard/settings`	`tenants:update`
System	`/dashboard/modules`	`tenants:update`
Customer	`/dashboard/pending-tenant`	Customer awaiting org approval
Public	`/join`	Customer register + join
Public	`/invite/[token]`	Accept staff or customer invite

Analytics sub-routes

ROUTE	FOCUS
`/dashboard/analytics/appointments`	Scheduling metrics
`/dashboard/analytics/inventory`	Stock metrics
`/dashboard/analytics/campaigns`	Campaign performance
`/dashboard/analytics/seo`	SEO scan metrics

Inventory sub-routes

ROUTE	FOCUS
`/dashboard/inventory/items`	Item catalog
`/dashboard/inventory/categories`	Categories
`/dashboard/inventory/movements`	Movement audit trail
`/dashboard/inventory/alerts`	Stock alerts
`/dashboard/inventory/requests`	Restock workflow
